

DET-012 | RDP Lateral Movement to Multiple Hosts

ID: DET-012

MITRE ID: T1021.001

Data Source: SecurityEvent

Severity: High

Tactic:Lateral Movement

Category:LATERAL MOVEMENT

ADX MAPPING LOGIC:

Mapping: Source = 'Attacker account', State = 'target hostname', StartTime binned = 'login time window'. We use dcount() with a threshold and datetime_diff() to measure speed of movement across hosts.

OBJECTIVE: Practice dcount() thresholds and datetime_diff() — operators for detecting lateral movement speed. Real Sentinel: SecurityEvent EventID 4624 LogonType 10 with one account authenticating to many hosts in a short window.

KQL QUERY (RUNNABLE ON ADX FREE TIER)

StormEvents

| where StartTime > ago(365d)

| where EventType in (

"Thunderstorm","Lightning","Heavy Rain") // = RDP logon events

| summarize

TargetHosts = dcount(State),

HostList = make_set(State, 10),

TotalLogins = count(),

FirstSeen = min(StartTime),

LastSeen = max(StartTime)

by Source, bin(StartTime, 1h)

| where TargetHosts > 3 // moved to >3 hosts = lateral movement

| extend

SpeedMins = datetime_diff('minute', LastSeen, FirstSeen),

AvgHostsPerMin = round(todouble(TargetHosts) /

(datetime_diff('minute',LastSeen,FirstSeen)+1), 2),

RiskLevel = iff(TargetHosts > 8, "CRITICAL",

iff(TargetHosts > 5, "HIGH", "MEDIUM"))

| project Source, TargetHosts, TotalLogins,

SpeedMins, AvgHostsPerMin, RiskLevel,HostList, FirstSeen, LastSeen

| order by TargetHosts desc

| take 20

INVESTIGATION STEPS

1. Verify if the account belongs to a sysadmin with legitimate access.
2. Check ServiceNow for a change ticket authorising the activity.
3. Review what was done on each host after RDP login.
4. Check if the source IP is internal or an external attacker.

RESPONSE ACTIONS

- If unauthorised: disable the account immediately.
- Review all hosts in HostList for compromise indicators.
- Check for persistence on each host (tasks, services, registry).
- Escalate to the IR team if no valid change ticket is found.

Output from Above KQL Query:

```
1 StormEvents
2 | where StartTime > ago(365d)
3 | where EventType in ("Thunderstorm","Lightning","Heavy Rain") // = RDP logon events
4 | summarize
5   TargetHosts = dcount(State),
6   HostList = make_set(State, 10),
7   TotalLogins = count(),
8   FirstSeen = min(StartTime),
9   LastSeen = max(StartTime)
10 by Source, bin(StartTime, 1h)
11 | where TargetHosts > 3 // moved to >3 hosts = lateral movement
12 | extend
13   SpeedMins = datetime_diff('minute', LastSeen, FirstSeen),
14   AvgHostsPerMin = round(todouble(TargetHosts) /
15     (datetime_diff('minute', LastSeen, FirstSeen)+1), 2),
16   RiskLevel = iff(TargetHosts > 8, "CRITICAL",
17     iff(TargetHosts > 5, "HIGH", "MEDIUM"))
18 | project Source, TargetHosts, TotalLogins, SpeedMins, AvgHostsPerMin, RiskLevel,
19   HostList, FirstSeen, LastSeen
20 | order by TargetHosts desc | take 20
```

Table 1 + Add visual Stats 2026-06-02 06:25 (UTC) Done (0.597 s)

Source	TargetHosts	TotalLogins	SpeedMins	AvgHostsPerMin	RiskLevel	HostList	FirstSeen	LastSeen
No Rows To Show								

KQL Detection Query in Sentinel:

SecurityEvent

```
| where TimeGenerated > ago(1h)
| where EventID == 4624
| where LogonType == 10 // RemoteInteractive = RDP
| summarize
  TargetHosts = dcount(Computer),
  HostList = make_set(Computer),
  TotalLogins = count(),
  FirstSeen = min(TimeGenerated),
  LastSeen = max(TimeGenerated)
by TargetUserName, IPAddress
| where TargetHosts > 3
| extend SpeedMins = datetime_diff(
'minute', LastSeen, FirstSeen)
| project TargetUserName, IPAddress, TargetHosts,
TotalLogins, HostList, SpeedMins,
FirstSeen, LastSeen
| order by TargetHosts desc
```